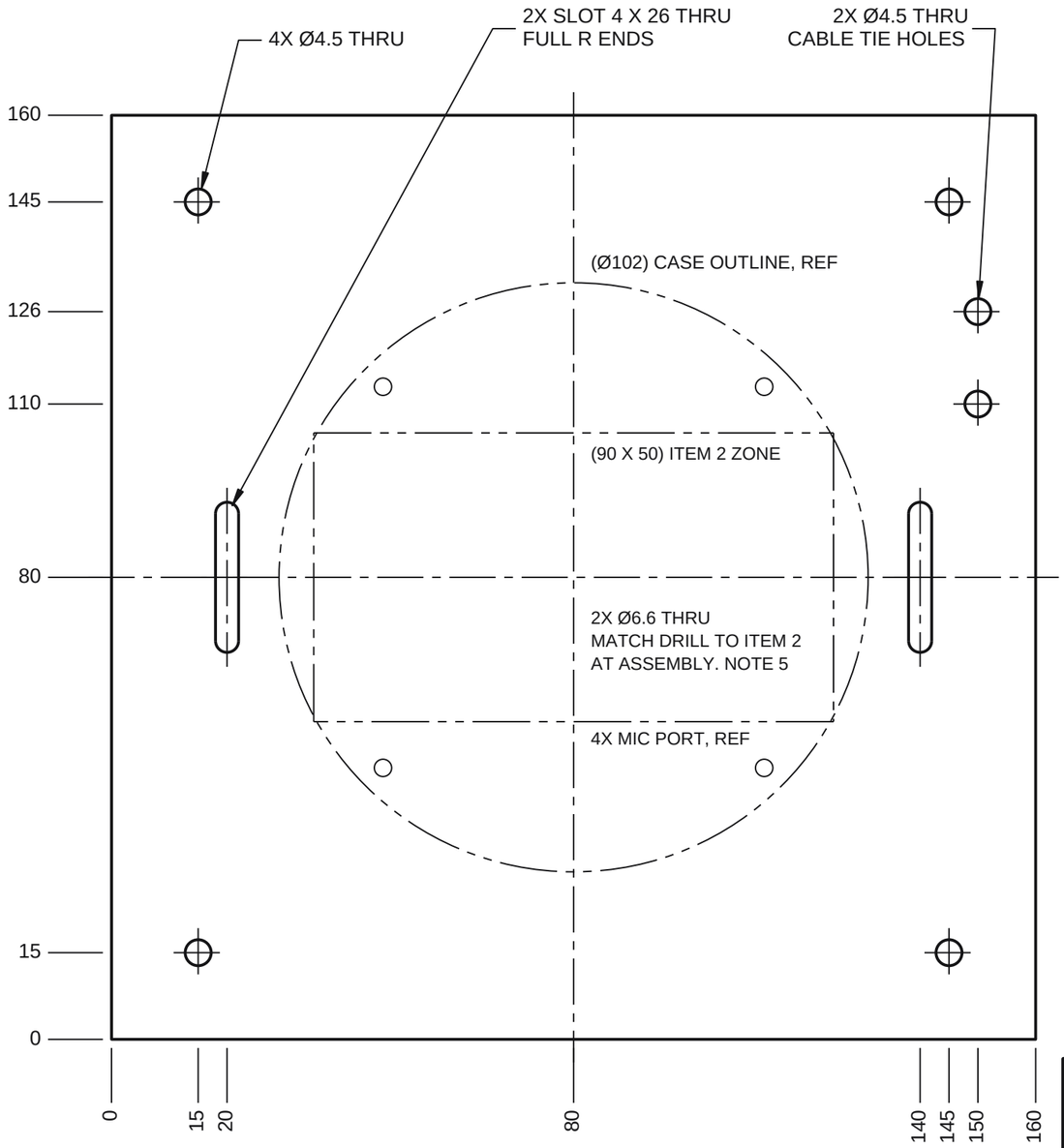
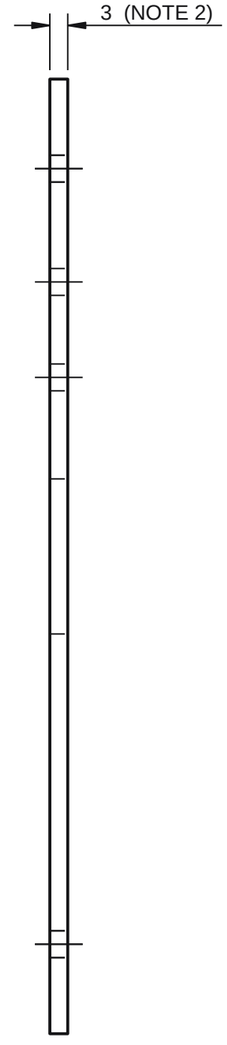
REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
A	INITIAL ISSUE. REPLACES REV B SKETCHES	2026-11-10	

FRONT VIEW

SCALE 1:1



RIGHT SIDE VIEW



NOTES:

1. INTERPRET DIMENSIONS AND TOLERANCES PER ASME Y14.5-2018.
2. MATERIAL: POLYCARBONATE SHEET, CLEAR, 3 THICK NOMINAL. 1/8 IN (3.2) STOCK IS ACCEPTABLE. McMASTER-CARR 8574K26 OR EQUIVALENT.
3. REMOVE ALL BURRS. BREAK SHARP EDGES 0.3 MAX.
4. ORDINATE DIMENSIONS ARE FROM THE LOWER LEFT CORNER OF THE FRONT VIEW.
5. AT ASSEMBLY, MATCH DRILL 2X Ø6.6 THRU INSIDE THE PHANTOM RECTANGLE TO TWO 1/4-20 UNC HOLES OF THE ADAPTER PLATE (SLM-PAS-A01 ITEM 2). ITEM 2 HOLE SPACING IS NOT VERIFIED.
6. PHANTOM CIRCLE AND MICROPHONE PORT POSITIONS ARE REFERENCE ONLY, FROM MANUFACTURER DATA. MEASURE THE PURCHASED CASE BEFORE CUTTING.
7. MARK +X AND +Y ON THE PART ONLY AFTER AXIS TEST HB-5 (docs/TEST_PROCEDURES.md).
8. CUTTING PROFILE: hardware/cad/carrier.dxf, LAYER CUT. 3D MODEL: hardware/cad/carrier.scad.
9. PRELIMINARY. THIS PART HAS NOT BEEN CUT OR FITTED.

UNLESS OTHERWISE SPECIFIED

DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
LINEAR ±0.5
HOLE DIAMETERS ±0.2
ANGLES ±1°
INTERPRET PER ASME Y14.5-2018

THIRD ANGLE PROJECTION

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		2026-11-10
CHECKED		
APPROVED		

MATERIAL
POLYCARBONATE SHEET, CLEAR, 3 THICK. SEE NOTE 2

FINISH
NONE. DEBURR, NOTE 3

STATUS PRELIMINARY

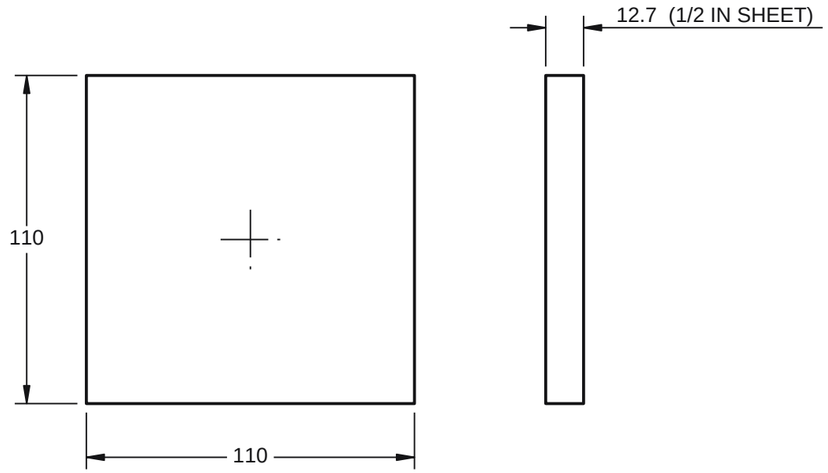
S&L MILESTONE SOLUTIONS

TITLE
CARRIER PLATE
PASSIVE ACOUSTIC SENSOR NODE, SENSOR HEAD

SIZE	DWG NO	REV
B	SLM-PAS-M01	A

SCALE	PROJECT	SHEET
1:1	PAS NODE	1 OF 1

REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
A	INITIAL ISSUE. REPLACES REV B SKETCHES	2026-11-10	

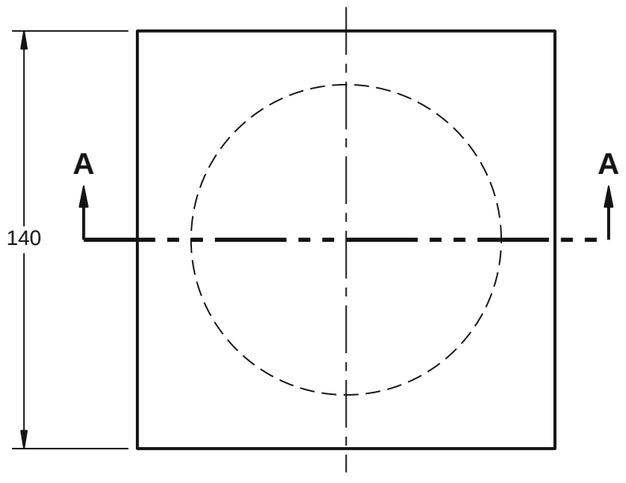


-1 ISOLATION PAD

SCALE 1:2. QTY 1

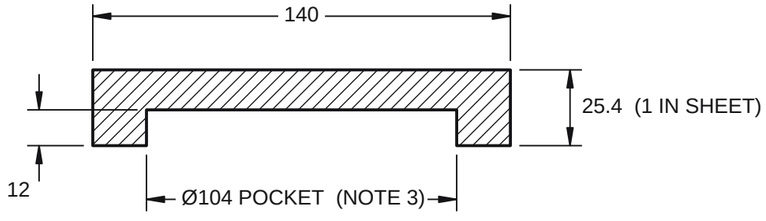
NOTES:

1. MATERIAL: RETICULATED (OPEN-CELL) POLYURETHANE FOAM, 30 PORES PER INCH. THE FOAM FACTORY FILTER FOAM OR EQUIVALENT. -1 FROM 1/2 IN SHEET, -2 FROM 1 IN SHEET.
2. CUT WITH A LONG SHARP BLADE. DO NOT COMPRESS THE FOAM WHILE CUTTING.
3. CUT THE POCKET IN -2 TO A LIGHT SLIP FIT OVER THE MEASURED CASE. Ø104 ASSUMES A Ø102 CASE, WHICH IS NOT VERIFIED.
4. THE POCKET IS BLIND. LEAVE AT LEAST 12 OF FOAM ABOVE THE MICROPHONE PORTS.
5. RELIEVE -1 LOCALLY OVER THE HEADS OF THE ASSEMBLY SCREWS IF THE CASE ROCKS (SLM-PAS-A01 NOTE 5).
6. FOAM IS PART OF THE ACOUSTIC PATH. REPEAT THE BEARING CALIBRATION (TEST HB-5) AFTER FITTING OR CHANGING -2.
7. PRELIMINARY. THESE PARTS HAVE NOT BEEN CUT OR FITTED.



-2 WINDSCREEN CAP, TOP VIEW

SCALE 1:2. QTY 1



SECTION A-A

SCALE 1:2

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
LINEAR ±2
HAND-CUT FOAM PARTS
INTERPRET PER ASME Y14.5-2018

THIRD ANGLE PROJECTION

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		2026-11-10
CHECKED		
APPROVED		

MATERIAL: RETICULATED POLYURETHANE FOAM, 30 PPI. SEE NOTE 1

FINISH: NONE

STATUS: **PRELIMINARY**

S&L MILESTONE SOLUTIONS

TITLE
FOAM PAD AND WINDSCREEN CAP
PASSIVE ACOUSTIC SENSOR NODE, SENSOR HEAD

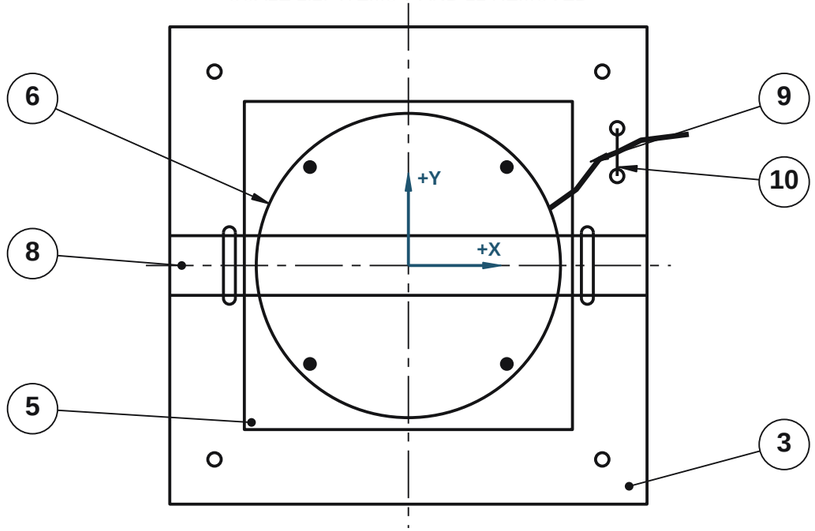
SIZE	DWG NO	REV
B	SLM-PAS-M02	A

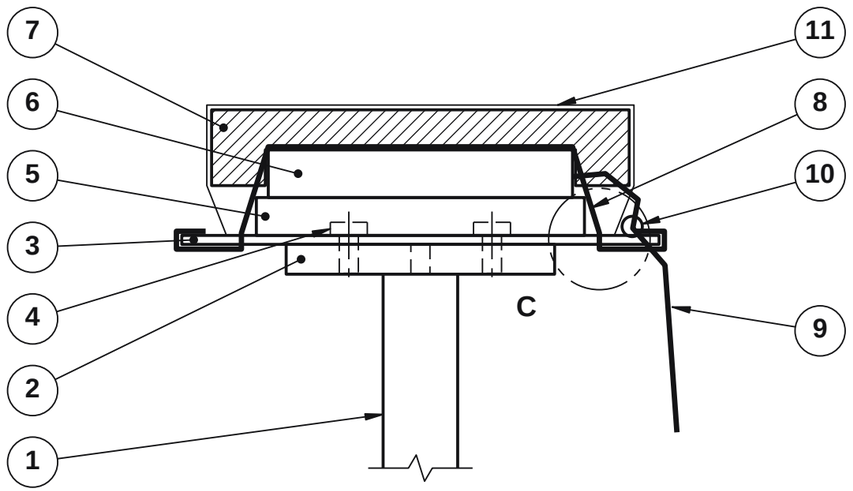
SCALE	PROJECT	SHEET
1:2	PAS NODE	1 OF 1

REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
A	INITIAL ISSUE. REPLACES REV B SKETCHES	2026-11-10	

TOP VIEW

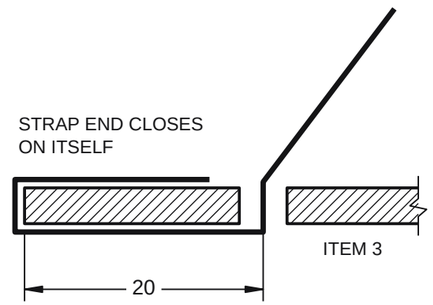
SCALE 1:2. ITEMS 7 AND 11 REMOVED





FRONT VIEW

SCALE 1:2. ITEM 7 IN SECTION



DETAIL C

SCALE 2:1. STRAP AT EACH SLOT

NOTES:

1. SEE hardware/HARDWARE_BUILD.md SECTION 6 FOR THE ASSEMBLY SEQUENCE AND INSPECTION POINTS.
2. MICROPHONE PORTS (LOGO SIDE OF ITEM 6) FACE UP. NOTHING SOLID ABOVE THE PORT PLANE.
3. ITEM 6 RESTS ON ITEM 5 ONLY. NO RIGID PART TOUCHES THE CASE.
4. STRAP ITEM 8 RUNS ALONG X BETWEEN THE MICROPHONE PAIRS. EACH END PASSES DOWN THROUGH A SLOT, AROUND THE PLATE EDGE, AND CLOSES ON ITSELF (DETAIL C). IT DOES NOT PASS UNDER THE CENTER OF THE PLATE, WHERE ITEMS 1 AND 2 SIT.
5. ITEM 4 SCREWS ENTER ITEM 2 FROM ABOVE THROUGH MATCH-DRILLED HOLES IN ITEM 3. SCREW POSITIONS SHOWN ARE NOT VERIFIED. RELIEVE ITEM 5 OVER THE SCREW HEADS IF THE CASE ROCKS.
6. FORM A LOOSE LOOP IN CABLE ITEM 9 AND TIE IT AT THE TWO TIE HOLES WITH ITEM 10.
7. ITEM 7 IS HELD BY TWO ELASTIC LOOPS, ITEM 11, RUN DIAGONALLY BETWEEN OPPOSITE CORNER HOLES.
8. LEVEL ITEM 3 WITHIN 2 DEGREES. WEIGHT THE BASE OF ITEM 1.
9. CASE HEIGHT, CASE OUTLINE, AND USB SOCKET POSITION ARE NOT VERIFIED. MEASURE THE PURCHASED UNIT.
10. PRELIMINARY. THIS ASSEMBLY HAS NOT BEEN BUILT.

PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	IMPACT LS-8AI	LIGHT STAND, 1/4-20 STUD
2	1	SMALLRIG 4887	ADAPTER PLATE, 90 X 50 X 10, 1/4-20 THREADS
3	1	SLM-PAS-M01	CARRIER PLATE
4	2	McMASTER-CARR 91772A537	SCREW, PAN HEAD, 1/4-20 X 1/2, 18-8 SST
5	1	SLM-PAS-M02-1	ISOLATION PAD, FOAM
6	1	SEEED 114993701	MICROPHONE ARRAY WITH CASE (A1)
7	1	SLM-PAS-M02-2	WINDSCREEN CAP, FOAM
8	1	STARTECH B506I-HOOK-LOOP-TIES	STRAP, HOOK AND LOOP, 3 TIES JOINED
9	1	STARTECH USB2AC2M	CABLE W1, USB-C TO USB-A, 2 m
10	2	COMMERCIAL	CABLE TIE, SMALL
11	2	COMMERCIAL	ELASTIC LOOP, CAP RETENTION
12	1	COMMERCIAL	SANDBAG OR WEIGHT FOR ITEM 1 (NOT SHOWN)

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN MILLIMETERS
REFERENCE DIMENSIONS IN ( )
NO TOLERANCED DIMENSIONS
ON THIS SHEET

THIRD ANGLE PROJECTION

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		2026-11-10
CHECKED		
APPROVED		

MATERIAL
SEE PARTS LIST

FINISH
NOT APPLICABLE

STATUS **PRELIMINARY**

S&L MILESTONE SOLUTIONS

TITLE
SENSOR HEAD ASSEMBLY
PASSIVE ACOUSTIC SENSOR NODE

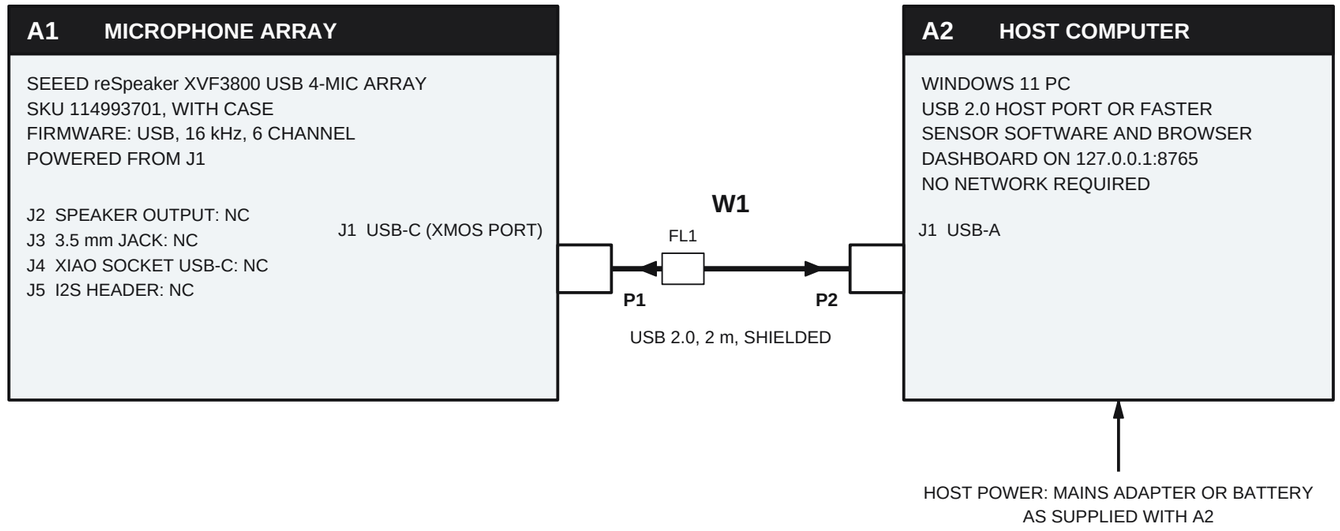
SIZE	DWG NO	REV
B	SLM-PAS-A01	A

SCALE	PROJECT	SHEET
1:2	PAS NODE	1 OF 1

REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
A	INITIAL ISSUE. REPLACES REV B SKETCHES	2026-11-10	

INTERCONNECTION

ONE CABLE. NO OTHER ELECTRICAL CONNECTIONS



NOTES:

1. W1 CONNECTS A1-J1 DIRECTLY TO A HOST PORT. NO UNPOWERED HUB.
2. TOTAL USB CABLE LENGTH 3 m MAX WITHOUT AN ACTIVE EXTENSION.
3. A1 IS POWERED FROM J1 ONLY. SUPPLY CURRENT IS NOT PUBLISHED BY THE MANUFACTURER. MEASURE IT (TEST HB-3). LIMIT 500 mA, ONE USB 2.0 PORT.
4. A1-J2 TO J5 ARE NOT CONNECTED (NC). NO SOLDERING. NO MAINS WIRING.
5. FL1 IS FITTED ONLY IF INTERFERENCE IS SEEN ON THE AUDIO. PLACE IT WITHIN 50 mm OF P1 AND TIE IT IN PLACE.
6. PIN ASSIGNMENTS ARE THOSE OF A STANDARD USB 2.0 TYPE-C TO TYPE-A CABLE. W1 IS A PURCHASED PART AND IS NOT MODIFIED.
7. CONNECTOR DESIGNATORS ON A1 ARE ASSIGNED BY THIS DRAWING. THEY ARE NOT THE MANUFACTURER'S MARKINGS.
8. THE FOUR RAW MICROPHONES ARE CAPTURE CHANNELS 2 TO 5, COUNTING FROM 0, OF THE 6 CHANNEL 16 kHz USB AUDIO STREAM. SEE firmware/PROVISIONING.md.
9. PRELIMINARY. NOT VERIFIED ON HARDWARE.

WIRE LIST, CABLE W1

CONDUCTOR	W1-P1 PIN (USB-C PLUG)	W1-P2 PIN (USB-A)	SIGNAL
1	A4, A9, B4, B9	1	VBUS, +5 V
2	A7	2	D-
3	A6	3	D+
4	A1, A12, B1, B12	4	GND
NONE	A5 (CC)	NO CONNECTION	56 KILOHM PULL-UP TO VBUS INSIDE P1
SHIELD	SHELL	SHELL	CABLE SHIELD, BONDED AT BOTH ENDS

REFERENCE DESIGNATIONS

REF	DESCRIPTION	PART NUMBER
A1	MICROPHONE ARRAY WITH CASE	SEEED 114993701
A2	HOST COMPUTER	EXISTING WINDOWS 11 PC
W1	CABLE, USB-C TO USB-A, 2 m	STARTECH USB2AC2M
FL1	FERRITE, SNAP-ON (OPTIONAL)	FAIR-RITE 0431164281

UNLESS OTHERWISE SPECIFIED
DIAGRAM. NOT TO SCALE
NO DIMENSIONS ON THIS SHEET

THIRD ANGLE PROJECTION

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		2026-11-10
CHECKED		
APPROVED		

MATERIAL
NOT APPLICABLE

FINISH
NOT APPLICABLE

STATUS **PRELIMINARY**

S&L MILESTONE SOLUTIONS

TITLE
INTERCONNECTION DIAGRAM
PASSIVE ACOUSTIC SENSOR NODE

SIZE	DWG NO	REV
B	SLM-PAS-E01	A

SCALE	PROJECT	SHEET
NONE	PAS NODE	1 OF 1